A Three-Element Monoid Counterexample to Weak Exactness of Hopf–von Neumann Preduals

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, for human review. It gives a negative answer to Question 17 of Douglas–Nowak [1]: the predual of a finite-dimensional, commutative Hopf–von Neumann algebra need not be weakly exact in the sense defined in that paper.

1 Source question

The source is Ronald G. Douglas and Piotr W. Nowak, *Every finitely generated group is weakly exact*, arXiv:1109.0313 [1]. In Section III, printed page 10, Definitions 15–16 introduce Hopf A-bimodules and call the predual $A = M_*$ of a Hopf–von Neumann algebra weakly exact when

$$\mathcal{H}^1(A, \mathcal{E}) = 0$$

for every weak-$*$ operator closed Hopf A-bimodule of the displayed form. The paper then asks:

Question 17. Is every Banach algebra A as above weakly exact?

The source page, including the definitions, module actions, and question, is shown below.

III. Weakly exact Banach algebras. The above results on bounded cohomology of groups suggest one might define a notion of weak exactness for certain Banach algebras. Such algebras have to be co-algebras in an appropriate sense, so that their duals are Banach algebras in a natural way as well. This requirement is a consequence of the fact that we have used the structure of $\ell_1(G)$ as a Hopf algebra, not only as a Banach algebra. We will consider only preduals of von Neumann algebras but it is clear that the definition can be extended to other cases.

Let M be a Hopf-von Neumann algebra and let $A = M_*$ denote a predual Banach algebra. Let X be a right A-module and consider the space $\mathcal{L}(X, M)$. The algebra M is an A-bimodule in a natural way, as it is the dual of the A-bimodule A. Thus $\mathcal{L}(X, M)$ is an A-bimodule with the following actions:

$$\begin{aligned}(a \cdot T)(x) &= T(xa),\\ (T \cdot a)(x) &= T(x)a,\end{aligned}$$

for $a \in A$, $T \in \mathcal{L}(X, M)$ and $x \in X$. Since M is an algebra, there is the additional structure of an M-module on $\mathcal{L}(X, M)$ given by

$$(bT)(x) = b(T(x)),$$

for $b \in M$, $T \in \mathcal{L}(X, M)$ and $x \in X$.

Definition 15. Let M be a Hopf-von Neumann algebra and A its predual Banach algebra. A submodule of $\mathcal{L}(X, M)$, which is both an A-bimodule and an M-module with respect to the structures described above, is called a Hopf A-bimodule.

Recall that given a Banach algebra A and an A-bimodule $\mathcal{E}$ one can define the Hochschild cohomology groups $\mathcal{H}^*(A, \mathcal{E})$ of A with coefficients in $\mathcal{E}$. In particular, the first cohomology group $\mathcal{H}^1(A, \mathcal{E})$ is defined as the quotient of the space of all A-derivations from A into X modulo the inner derivations, see for example [3, 10].

Definition 16. Let M be a Hopf-von Neumann algebra and let A be a predual Banach algebra of M. We define A to be weakly exact if

$$\mathcal{H}^1(A, \mathcal{E}) = 0$$

for every M-submodule $\mathcal{E} \subseteq \mathcal{L}(X, M)$, which is closed in the weak-* operator topology, where X is any left A-module.

It is natural to ask if dimension shifting preserves the class of Hopf modules over A and, more importantly, do algebras behave similarly to finitely generated groups:

Question 17. Is every Banach algebra A as above weakly exact?

References

[1] J. Brodzki, G.A. Niblo, P. Nowak, N. Wright, Amenable actions, invariant means and bounded cohomology, preprint arXiv:1004.0295. ↑1

Convention note. Immediately before Definition 15 the source takes X to be a *right* A-module and defines

$$(a \cdot T)(x) = T(xa), \qquad (T \cdot a)(x) = T(x)a. \tag{1}$$

Definition 16 later says “left A-module,” although (1) uses the right action. We use the displayed convention. Our X carries the same one-dimensional action on both sides, so it is also a left module if the later word is read literally.

2 Definitions and the finite monoid

Let $S = \{0, 1, 2\}$ with multiplication

$$\begin{array}{c|ccc} \cdot & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 2 \\ 2 & 0 & 2 & 2 \end{array}. \tag{2}$$

Thus 1 is the identity, and $st = t$ whenever $t \in \{0, 2\}$. Put

$$M = \ell_\infty(S), \qquad (\Delta f)(s,t) = f(st). \tag{3}$$

Then M is a finite-dimensional commutative von Neumann algebra. Its predual $A = M_* = \ell_1(S)$ has the convolution product

$$\delta_s \delta_t = \delta_{st}. \tag{4}$$

For the natural dual A-bimodule structure on M, we use

$$\langle h \cdot a, b\rangle = \langle h, ab\rangle \quad (a, b \in A,\ h \in M). \tag{5}$$

In particular,

$$(h \cdot \delta_s)(r) = h(sr). \tag{6}$$

A bounded linear map $D : A \to \mathcal{E}$ is a derivation when

$$D(ab) = a \cdot D(b) + D(a) \cdot b. \tag{7}$$

It is inner when it has the form $a \mapsto a \cdot \xi - \xi \cdot a$ for some $\xi \in \mathcal{E}$.

3 Proof intuition

The monoid (2) has two different characters on its semigroup algebra. The augmentation character φ is constantly one on the three point masses, while the coefficient-at-the-identity character ψ is one only at δ_1. The coordinate line in M supported at the identity is a Hopf A-bimodule whose left action is φ and whose right action is ψ.

Derivations into this line are therefore (φ, ψ)-derivations. The asymmetry in the right-zero ideal $\{0, 2\}$ lets a derivation distinguish 0 from 2: it takes values $(1, 0, 0)$ on $(\delta_0, \delta_1, \delta_2)$. Inner derivations cannot make that distinction; all of them have values proportional to $(1, 0, 1)$. Hence the first Hochschild cohomology is nonzero. All ambient spaces are finite dimensional, so the required operator weak-$*$ closedness is automatic.

4 Counterexample theorem

Theorem 1. *There is a finite-dimensional commutative Hopf–von Neumann algebra M whose predual Banach algebra $A = M_*$ is not weakly exact in the sense of Douglas–Nowak. Consequently, the answer to their Question 17 is negative.*

Proof. We verify each hypothesis and then exhibit a non-inner derivation.

1. The Hopf–von Neumann algebra. The table (2) is associative. Indeed, if the final factor u is 0 or 2, then both $(st)u$ and $s(tu)$ equal u; if $u = 1$, both equal st. The element 1 is visibly a two-sided identity.

It follows that the map Δ in (3) is a unital $*$-homomorphism and is coassociative. It is normal because M is finite dimensional. It is also injective: if $\Delta f = 0$, then

$$f(t) = (\Delta f)(1,t) = 0 \qquad (t \in S).$$

Thus the example satisfies even the convention in which injectivity is included in the definition of a Hopf–von Neumann coproduct.

2. Two characters. For $a = \sum_{s\in S} a_s\delta_s \in A$, define

$$\varphi(a) = a_0 + a_1 + a_2, \qquad \psi(a) = a_1. \tag{8}$$

The augmentation φ is multiplicative. The functional ψ is multiplicative as well, because $st = 1$ in S if and only if $s = t = 1$.

Let $X = \mathbb{C}$ with $x \cdot a = \varphi(a)x$; give it the same left action as noted above. Identify $\mathcal{L}(X, M)$ with M via $T \leftrightarrow T(1)$. Let

$$e = \mathbf{1}_{\{1\}} \in M, \qquad \mathcal{E} = \mathbb{C}e. \tag{9}$$

3. $\mathcal{E}$ is an admissible Hopf A-bimodule. For the left action in (1),

$$a \cdot e = \varphi(a)e. \tag{10}$$

For the right action, (6) and the fact that $sr = 1$ only when $s = r = 1$ give

$$e \cdot a = \psi(a)e. \tag{11}$$

Hence $\mathcal{E}$ is an A-bimodule. It is also an M-module under pointwise multiplication, since $he = h(1)e$ for every $h \in M$. Finally $\mathcal{E}$ is finite dimensional, and hence is closed in the weak-$*$ operator topology. It is therefore a Hopf A-bimodule of exactly the type quantified over in Definition 16 of the source.

4. A derivation. Define $D : A \to \mathcal{E}$ on the point masses by

$$D(\delta_0) = e, \qquad D(\delta_1) = D(\delta_2) = 0. \tag{12}$$

Write $d(0) = 1$ and $d(1) = d(2) = 0$, so that $D(\delta_s) = d(s)e$. By (10)–(11), the derivation identity on point masses is

$$d(st) = d(t) + d(s)\psi(\delta_t). \tag{13}$$

If $t = 0$, both sides of (13) are 1; if $t = 2$, both are 0; and if $t = 1$, both are $d(s)$. Thus (13) holds for every $s, t \in S$. Bilinearity proves (7) for arbitrary $a, b \in A$. The map D is bounded because A and $\mathcal{E}$ are finite dimensional.

5. The derivation is not inner. Every $\xi \in \mathcal{E}$ is λe for some $\lambda \in \mathbb{C}$. By (10)–(11), the corresponding inner derivation has point-mass values

$$\lambda\big(\varphi(\delta_s) - \psi(\delta_s)\big)e, \qquad s = 0, 1, 2,$$

that is, the coefficient vector

$$\lambda(1, 0, 1). \tag{14}$$

The derivation (12) has coefficient vector $(1, 0, 0)$, which is not of the form (14). Therefore D is not inner, so

$$\mathcal{H}^1(A, \mathcal{E}) \neq 0.$$

The predual A is not weakly exact. $\square$

5 Verification and discovery audit

The proof above is independent of computation. The included direct verifier checks all 27 associativity identities, the identity element, surjectivity of multiplication (and hence injectivity of Δ), multiplicativity of φ and ψ, all nine basis instances of (13), and the incompatible inner/derivation vectors. It exits successfully.

The example was found by the included finite-semigroup search, which uses exact rational linear algebra. The exhaustive order-two run checks all 16 tables, finds 8 associative ones and 14 stable coordinate-module cases, and finds no nonzero cohomology. The order-three run checks all $3^9 = 19683$ tables, finds 113 associative tables and 341 stable coordinate-module cases, with 27 hits. For the unital example (2) with support $\{1\}$ it computes

$$\dim \mathrm{Der}(A, \mathcal{E}) = 2, \quad \dim \mathrm{Inn}(A, \mathcal{E}) = 1, \quad \dim \mathcal{H}^1(A, \mathcal{E}) = 1.$$

This enumeration is a discovery and sanity-check aid, not a premise of the proof.

6 Novelty bounds, limitations, and review recommendation

On 9 August 2026, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:1109.0313 and the core phrases. The local full-source arXiv corpus and exact official-arXiv searches for *"weakly exact Banach algebra," "Is every Banach algebra weakly exact,"* and *"Hopf A-bimodule weakly exact"* found the source paper but no later answer to Question 17. Within those bounded searches, this appears to be a new counterexample; publication-level novelty is not certified.

The theorem answers the broad question under the Hopf–von Neumann and Hopf module definitions printed in the source. The example is finite dimensional, unital, commutative, and has an injective coproduct. It is not a group algebra and is not claimed to satisfy the extra antipode, cancellation, Haar-weight, or locally compact quantum-group axioms that some narrower frameworks impose.

Human review should focus on (i) the orientation of the natural dual action in (5)–(6), (ii) the source's right/left-module wording inconsistency and the fact that X carries both actions here, and (iii) whether the intended scope of "Hopf–von Neumann algebra" in Question 17 silently includes any axioms beyond a normal unital coassociative coproduct. No such extra axioms are stated in the question's setup, and injectivity is verified explicitly.

References

[1] R. G. Douglas and P. W. Nowak, *Every finitely generated group is weakly exact*, arXiv:1109.0313, Section III, Definitions 15–16 and Question 17, `https://arxiv.org/abs/1109.0313`.